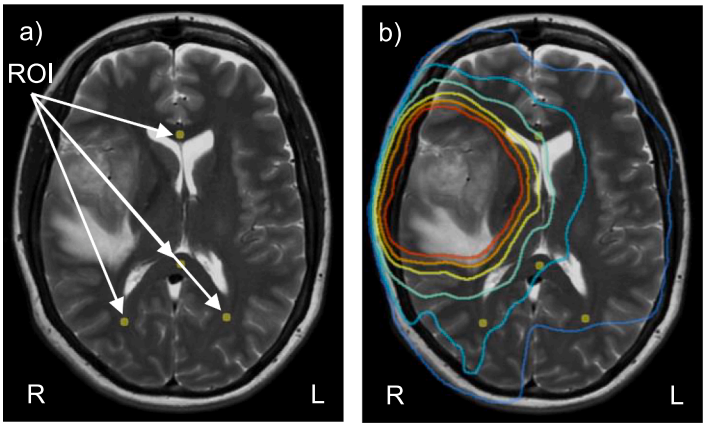


Fig. 2. Synthetic T_2-weighted image with ROI and dose distribution. In a) there is an example of ROI placements in a synthetic T_2-weighted image of a patient. ROIs are placed in the lower part of the parietal lobe (left and right), in the genu and splenium of the corpus callosum. In b) the same T_2-weighted image is displayed, but with an overlay of the isodose lines of 10, 20, 30, 40, 50 and 57 Gy.

Table 2
Student's *t*-test of differences from baseline for immediate post-operative examination and 1st follow-up after RT. There was no significant difference between baseline and the first post-operative MRI. Significant changes in PD, c_{My} and R_2 were observed during the first follow-up after RT completion. The difference is expressed as the arithmetic difference between the two percentages, percent point (p.p.), for PD and c_{My} and in per second for the relaxation rates.

	1st post op. mean of difference *(n = 104)*	p-value	1st follow-up mean of difference *(n = 116)*	p-value
PD (p.p.)	0.32	ns	1.5	<0.001
c_{My}(p.p.)	−0.55	ns	−2.2	<0.001
R_1 (s^{-1})	−0.008	ns	−0.013	ns
R_2 (s^{-1})	−0.11	ns	0.33	<0.001

mm slice thickness and 1 mm gap, were collected with a scan time of 5:55 min. The qMRI maps (R_1, R_2, and PD) and the myelin map (c_{My}) were calculated and synthetic T_2-weighted images reconstructed. In Fig. 1 R_1, R_2, PD and c_{My} are shown for one patient from the examinations before and after operation and the first three follow-up examinations after RT completion.

2.4. Data analysis

A total of 83 MRI examinations were concluded for the ten patients during a two-year follow-up period. Out of these, ten were pre-operative examinations (baseline), nine were radicality controls after surgery (1st post op.) and 57 were conducted after CRT (follow-up). On seven occasions extra examinations were done between the radicality control and CRT. These were excluded, leaving 76 examinations for analysis. The number of follow-up examinations varied between two to nine, the follow-up period was up to two years or until death. Circular ROIs with a diameter of 5 mm, 4 mm if the anatomical structure was less than 5 mm (less than 5% of the ROIs), were placed in NAWM in the synthetic T_2-weighted images by a neuroradiologist (IB). ROI placement was done in the MICE toolkit (vers. 2021.1.0, NONPI Medical AB, Sweden, Umeå). Fig. 2 shows an example of ROIs drawn in one slice. ROIs were placed in up to 14 anatomical regions: the genu and splenium of the corpus callosum, bilaterally in the pedunculus cerebellaris, peritrigonally and in a lower and upper slice of the frontal and parietal lobes. All ROIs were manually placed in a consistent manner to compensate for anatomical shifts in all examinations. However, ROIs were not drawn if the radiologist assessed that an oedema was present in the specific tissue area during the follow-up, or if there was tumour present at baseline. In Appendix A Supplementary data 1, Fig. A.1 displays a histogram of the number of ROIs for each time period and Table A.1 includes the number of ROIs in the different anatomical regions.

The absorbed dose distribution and CT-image were exported from Eclipse and transformed to the qMRI-space by performing a rigid registration of the CT-images to the synthetic T_2-volume, using the MICE toolkit. For every ROI a mean value of R_1, R_2, PD, c_{My} and absorbed dose was extracted, relaxation rates are expressed as per second, PD and c_{My} are expressed in percent and the absorbed dose in Gray. Using Matlab (vers. R2020B, The MathWorks Inc., Natick, MA, USA) the difference in one ROI compared with its baseline value was calculated for all ROIs and time-points. For PD and c_{My} the difference was expressed as the arithmetical difference of percent, percent points (p.p.). Appendix A Supplementary data 2 is an Excel sheet containing all ROIs and their quantitative values and the differences compared with baseline, used for the statistical tests performed.

2.5. Statistical tests

Statistical analysis was performed with SPSS Statistics v.28 (IBM Corp, Armonk, USA).

A two-sided paired Student's *t*-test was used to analyse changes in any quantitative value for all ROIs between baseline and the first follow-up after CRT. Another paired *t*-test evaluated the differences between the pre-operative and the first post-operative images taken for radicality control in order to exclude changes prior to CRT. The paired Student's *t*-test was used to exclude changes depending on patient, age, gender and anatomical location.

Changes related to absorbed dose and time after concluded CRT were analysed with a mixed model analysis. In this test, each ROI for every patient was seen as a subject in the analysis, with a repeated measurement at approximately every third month. The dependent variable (the difference in R_1, R_2, PD or c_{My}) were assumed to be linearly dependent on the covariates absorbed dose (Gy) and time (months), both of which were assumed to be fixed effects. Random effects depending on patient and anatomical location were accounted for in the mixed model. Significance level was set to 0.017, to correct for multiple comparison of three different statistical tests.

Table 3
The covariate factors and the significance of the mixed model test, the time factor is per month and the dose factor per Gray. PD and c_{My} are expressed as 10^{-3} and the arithmetic difference between the two percentages, percent point (p.p.) and the relaxation rates are expressed as 10^{-3} per second.

	Time factor ($month^{-1}$)	p-value	Dose factor (Gy^{-1})	p-value
PD (10^{-3}p.p.)	−42	0.005	26	<0.001
c_{My} (10^{-3}p.p.)	67	0.002	−42	<0.001
R_1 ($10^{-3}s^{-1}$)	2	<0.001	0	ns
R_2 ($10^{-3}s^{-1}$)	−17	<0.001	7	<0.001

3. Results

Comparisons of the differences in R_1, R_2, PD and c_{My} between the baseline image and the first follow-up show a significant difference in all parameters except R_1. The mean value of difference in R_1 is – 0.013 s^{-1} (p-value 0.13). For R_2 the difference is 0.33 s^{-1}, for PD it is 1.5 percentage points (pp) and the difference in c_{My} is −2.2 pp, all with a p-value less than 0.001. There was no significant difference between the pre-operative and radicality control images. The mean values of these differences are printed in Table 2.

The results of the mixed model test are printed in Table 3. There is a significant correlation with time and absorbed doses for all parameters except R_1 where only time is significant. For an absorbed dose of 50 Gy, the contribution to the difference of R_2 corresponded to 0.35 s^{-1}, PD 1.3 p.p. and c_{My} decreased by 2.2 p.p. The effect of time in the model corresponded to a difference in R_1 of 0.024 s^{-1}, R_2 −0.20 s^{-1}, PD −0.50 p.p. and c_{My} 0.80 p.p at 12 month.

A visual presentation of the mean values of change from baseline and the 95% confidence interval for mean value of ROIs difference for all quantitative parameters is found in Fig. 3. In the plot there is a distinction between ROIs receiving low or high doses with a threshold of 30 Gy.

4. Discussion

The aim of this work was to conclude if qMRI could be useful for detecting changes in NAWM in brain tumour patients after CRT. For this purpose two different statistical test were performed on the quantitative data that had been acquired. The results of the Student's *t*-tests confirm that differences can be detected and that the changes did not occur due to surgery but due to CRT. The result concluded no significant changes during the radicality control although significant differences were found

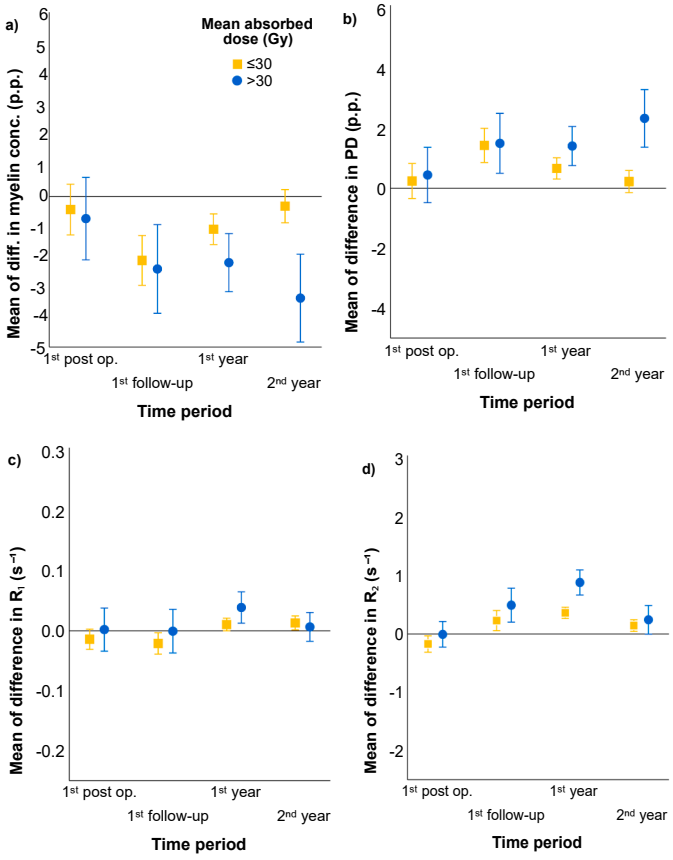


Fig. 3. Mean difference for different time period and absorbed dose. Mean differences for doses less (yellow square) or more (blue circle) than 30 Gy for different examination times. In panel a) the difference in percent points in myelin concentration is displayed, b) contains the difference of percent points in proton density, and c) and d) are the difference per second of relaxation rates R_1 and R_2. The error bars indicate the 95% confidence interval for mean value of ROIs differences. (i) A decrease of 2% was observed for c_{My}, during the first follow-up. This was then returned to baseline for low doses, but not for higher doses during the follow-up period. (ii) An increase in PD was observed during the first follow-up, followed by a subsequent return to baseline for low doses. (iii) In R_1 there were very small differences for all doses regardless of time-point. (iv) In R_2 an increase was detected, with a maximum difference during the first year, before returning to baseline. For higher doses a larger increase in R_2 was observed.